SIMATS Online Programming Lab

JAVA PROGRAMMING

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192421063RD

QUESTIONS 8 / 8 completed

Logger

LOGGER

You are working on a project that requires the use of a logging framework. Write a Java program to create an interface called "Logger" with methods for logging different types of messages, such as "debug", "info", "warning", and "error". Implement the interface in a class called "Log4jLogger" that uses the Log4j logging framework.

PROGRAM EDITOR Java Run Save

```
    public void info(String msg) {
        System.out.println("[INFO] " + msg);
    }

    public void warning(String msg) {
        System.out.println("[WARNING] " + msg);
    }

    public void error(String msg) {
        System.out.println("[ERROR] " + msg);
    }

    public static void main(String[] args) {
        Log4jLogger logger = new Log4jLogger();

        logger.debug("Debug message");
        logger.info("Information message");
        logger.warning("Warning message");
        logger.error("Error message");
    }
}
```

OUTPUT

```
[DEBUG] Debug message
[INFO] Information message
[WARNING] Warning message
[ERROR] Error message
```

INPUT

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QUESTIONS 8 / 8 completed

Character

CHARACTER

You are developing a game that involves different types of characters, such as heroes, villains, and monsters. Write a Java program to create an interface called "Character" with methods for moving, attacking, and taking damage. Implement the interface in classes for each type of character, such as "Hero", "Villain", and "Monster".

PROGRAM EDITOR Java Run Save

```java
interface Character {
    void move();
    void attack();
    void takeDamage(int damage);
}

public class GameCharacter implements Character {

    public void move() {
        System.out.println("Character is moving.");
    }

    public void attack() {
        System.out.println("Character is attacking.");
    }

    public void takeDamage(int damage) {
        System.out.println("Character took " + damage + " damage
    }

    public static void main(String[] args) {
```

OUTPUT

```
Character is moving.
Character is attacking.
Character took 25 damage.
```

INPUT

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QUESTIONS 8 / 8 completed

Sort

SORT

Write a Java program to create a list of integers and sort them in ascending order using the Collections framework.

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```
import java.util.*;

public class Main {
    public static void main(String[] args) {

        ArrayList<Integer> list = new ArrayList<>();

        list.add(50);
        list.add(20);
        list.add(40);
        list.add(10);
        list.add(30);

        Collections.sort(list);

        System.out.println("Sorted List: " + list);
    }
}
```

OUTPUT

```
Sorted List: [10, 20, 30, 40, 50]
```

INPUT

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QUESTIONS 8 / 8 completed

HashSet

HASHSET

Write a Java program to create a HashSet of strings and remove a specific element from it.

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```java
import java.util.HashSet;

public class Main {
    public static void main(String[] args) {

        HashSet<String> set = new HashSet<>();

        set.add("Apple");
        set.add("Banana");
        set.add("Orange");
        set.add("Mango");

        System.out.println("Before removing: " + set);

        System.out.println("After removing: " + set);
    }
}
```

OUTPUT

```
Before removing: [Apple, Mango, Orange, Banana]
After removing: [Apple, Mango, Orange, Banana]
```

INPUT

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QUESTIONS 8 / 8 completed

TreeMap

TREEMAP

Write a Java program to create a TreeMap of strings and integers and iterate over its entries.

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```java
import java.util.*;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        TreeMap<String, Integer> map = new TreeMap<>();

        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 0; i < n; i++) {
            String key = sc.nextLine();
            int value = sc.nextInt();
            sc.nextLine();
            map.put(key, value);
        }

        for (Map.Entry<String, Integer> entry : map.entrySet())
            System.out.println(entry.getKey() + " = " + entry.ge
```

OUTPUT

```
Apple = 10
Banana = 20
Mango = 40
Orange = 30
```

INPUT

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JAVA PROGRAMMING

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QUESTIONS 8 / 8 completed

PriorityQueue

PRIORITYQUEUE

Write a Java program to create a PriorityQueue of integers and add elements to it.

PROGRAM EDITOR Java Run Save

```
import java.util.PriorityQueue;

public class Main {
    public static void main(String[] args) {

        PriorityQueue<Integer> pq = new PriorityQueue<>();

        pq.add(50);
        pq.add(20);
        pq.add(40);
        pq.add(10);
        pq.add(30);

        System.out.println("Priority Queue: " + pq);
    }
}
```

OUTPUT

```
Priority Queue: [10, 20, 40, 50, 30]
```

INPUT

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QUESTIONS

8 / 8 completed

LinkedHashMap

LINKEDHASHMAP

Write a Java program to create a LinkedHashMap of strings and integers and retrieve the keys in the order they were inserted.

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PROGRAM EDITOR

Java

Run Save

```java
import java.util.*;

public class Main {
    public static void main(String[] args) {

        LinkedHashMap<String, Integer> map = new LinkedHashMap<>()

        map.put("Apple", 10);
        map.put("Banana", 20);
        map.put("Orange", 30);
        map.put("Mango", 40);

        System.out.println("Keys in insertion order:");

        for (String key : map.keySet()) {
            System.out.println(key);
        }
    }
}
```

OUTPUT

```
Keys in insertion order:
Apple
Banana
Orange
Mango
```

INPUT

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QUESTIONS 8 / 8 completed

Max-Min

MAX-MIN

Write a Java program to create a Set of integers and perform different operations on it, such as finding the maximum and minimum values.

PROGRAM EDITOR Java Run Save

```
import java.util.*;

public class Main {
    public static void main(String[] args) {

        Set<Integer> set = new HashSet<>();

        set.add(50);
        set.add(20);
        set.add(40);
        set.add(10);
        set.add(30);

        System.out.println("Set: " + set);
        System.out.println("Maximum value: " + Collections.max(set
        System.out.println("Minimum value: " + Collections.min(set
    }
}
```

OUTPUT

```
Set: [50, 20, 40, 10, 30]
Maximum value: 50
Minimum value: 10
```

INPUT